AFS-Net: An Anchor-Free Domain Adaptation Network for SAR Ship Detection

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***Abstract*—Object detection in synthetic aperture radar (SAR) imagery is crucial for maritime surveillance, yet it is often limited by the scarcity of large-scale annotated data. To address this challenge, this work proposes AFS-Net, a novel anchor-free domain adaptation framework for SAR ship detection. AFS-Net effectively transfers knowledge from easily annotated optical source-domain images to unlabeled SAR target-domain images, mitigating the need for expensive SAR data annotation. Unlike conventional anchor-based detectors that struggle with the diverse scales and shapes of ships, our anchor-free approach, built upon CenterNet, provides a more streamlined and accurate localization paradigm. The core of AFS-Net consists of two innovative alignment modules designed to counteract domain shifts: the keypoint alignment module (KAM) and the box alignment module (BAM). Specifically, KAM aligns the structural distribution of heatmap responses across domains, forcing the target-domain features to mimic the ideal Gaussian-like keypoint structures from the source domain, thereby enhancing keypoint localization robustness. Concurrently, BAM aligns the center offset and size regressions to ensure the geometric consistency of the predicted bounding boxes. Experimental results show that the proposed AFS-Net outperforms existing domain adaptation object detection frameworks on SAR ship detection tasks, achieving AP improvements of 12.10%–33.37% over the baseline on cross-domain settings.**

***Index Terms*—Anchor-free, domain adaptation, keypoint alignment, synthetic aperture radar (SAR) ship detection.**

I. Introduction

SYNTHETIC aperture radar (SAR) is capable of avoiding signal attenuation caused by weather conditions and can capture image information day and night [1]. As a result, SAR image object detection has received increasing attention, with ship detection being one of its most important applications.

Deep learning-based object detection algorithms require a sufficiently large annotated dataset for training [2], [3], [4], [5]. However, due to the unique imaging mechanism of SAR, its images generally have lower spatial resolution compared to optical images and are often affected by significant speckle noise and artifacts, which pose challenges for large-scale annotation.

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While the low-level imaging physics of optical and SAR imagery differ, they often share high-level semantic information, such as the geometric structure and spatial layout of objects, like ships. This shared semantic space makes knowledge transfer via domain adaptation a feasible and promising approach.

To address this limitation, we can utilize easily accessible labeled optical datasets as the source domain and unlabeled SAR datasets as the target domain, employing unsupervised domain adaptation algorithms for SAR ship detection.

In recent years, most domain adaptation methods from optical to SAR images have been anchor-based. For instance, some methods employed multilevel feature alignment across different representation granularities to reduce domain discrepancies [6], [7], [8]. Pan et al. [9] proposed an imbalanced discriminant alignment to enhance feature discriminability and prevent negative migration. Additionally, techniques such as pseudo-labeling and self-training were leveraged to adapt the model to the target domain [10], [11].

However, in the ship detection domain adaptation from optical to SAR images, anchor-based methods face inherent challenges. The extreme aspect ratios and scale variations of ships often lead to low initial IoU with predefined anchors, which strains the regression module and compromises localization accuracy. In contrast, anchor-free detectors like CenterNet [12] reformulate the task as keypoint estimation: they first locate the ship's center and then regress the bounding box (BBox). This approach avoids preset anchor shapes, making it inherently more adaptable to large morphological changes and thus providing a more robust foundation for domain adaptation. Recent anchor-free domain adaptation methods have emerged, such as the one by Zhang et al. [13] that suppresses background interference via foreground-enhanced multigranularity alignment. AFS-Net explores a more direct pathway by performing alignment in the model's output space, leveraging its high-level semantic abstraction and inherent task relevance for superior adaptation.

This letter proposes a novel anchor-free domain adaptation framework for SAR ship detection (AFS-Net). The framework comprises two main modules: the keypoint alignment module (KAM), which aligns the heatmaps of the source and target domains to leverage the structural information of ships, and the box alignment module (BAM), which aligns size maps and offset maps to further reduce domain gaps. KAM enables the detector to learn local Gaussian-distributed features of keypoint responses, thereby improving its ability to select keypoints in the target domain. BAM enhances the model's capability to represent and discriminate geometric structures.

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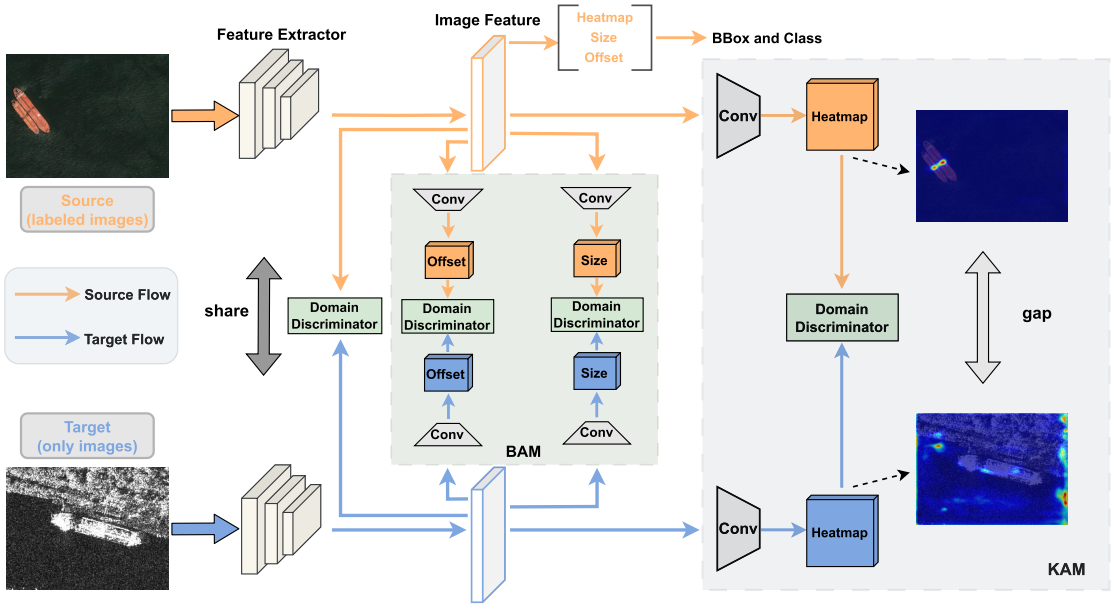


Fig. 1. Overview of our proposed domain adaptation framework. The model is trained with labeled source data for supervised object detection and simultaneously on unlabeled target data for adaptation. To bridge the domain gap, we propose two complementary alignment modules: KAM, which aligns the heatmap distributions to ensure robust localization, and BAM, which aligns offset and size predictions to achieve geometric consistency.

In summary, the contributions of this letter are as follows.

1) A novel anchor-free domain adaptation algorithm is introduced for SAR ship object detection.
2) To capture the domain-invariant structural features, we align the heatmaps of the source and target domains, enabling the model to learn stable keypoint distributions across modalities.
3) Adaptation is further enhanced through Box Alignment, which aligns keypoint offsets and BBox sizes to achieve comprehensive domain alignment.

II. Methodology

This section provides an overview of the framework, the KAM, the BAM, and the objective function of the proposed methodology.

We define the annotated optical images, referred to as the source domain, as $D_s = \{(x_i, y_i)\}_{i=1}^{n_s}$, where x_i denotes the ith optical image. y_i represents its annotation and n_s is the total number of optical images. Similarly, the unannotated SAR images, referred to as the target domain, are defined as $D_t = \{x_i\}_{i=1}^{n_t}$, where x_i denotes the ith image and n_t is the total number of samples in the target domain.

A. Framework Overview

The framework of AFS-Net is illustrated in Fig. 1. It is built upon the anchor-free CenterNet architecture and introduces adversarial learning for comprehensive output-space alignment. Given source and target images, the shared backbone first extracts feature maps. The detection head then generates three outputs: a heatmap indicating the probability of target centers, an offset map for subpixel correction, and a size map estimating object width and height.

To mitigate the domain gap, two complementary alignment modules are designed. KAM aligns heatmaps across domains, leveraging the structural consistency of ships to ensure accurate localization. BAM applies adversarial alignment to the offset and size maps, promoting invariance in fine-grained localization and object scale estimation.

Adversarial learning is implemented using gradient reversal layers (GRLs) [14]. During backpropagation, the GRL reverses the gradient direction, enabling the backbone network to learn domain-invariant features that confuse the domain discriminators. Dedicated discriminators are deployed for each alignment task, enabling the network to learn invariant representations at multiple levels. Each discriminator is a lightweight convolutional network with a similar architecture, tailored to its specific input tensor.

Meanwhile, standard classification and regression losses are applied to source samples to retain strong detection performance during training.

B. Keypoint Alignment Module

Heatmaps represent the keypoint predictions for both the source and target domains, capturing the spatial distribution of target center points.

As illustrated in Fig. 2, the KAM module is designed to rectify localization inaccuracies by aligning the structural properties of the heatmaps. In the CenterNet framework, the ground-truth heatmap for each object is a Gaussian kernel, which represents an ideal, unambiguous localization signal: a single, sharp peak at the object's center. This structure effectively maximizes the signal-to-noise ratio for the final keypoint detection. However, when a model trained on clean optical images is applied to SAR data, it often produces diffuse

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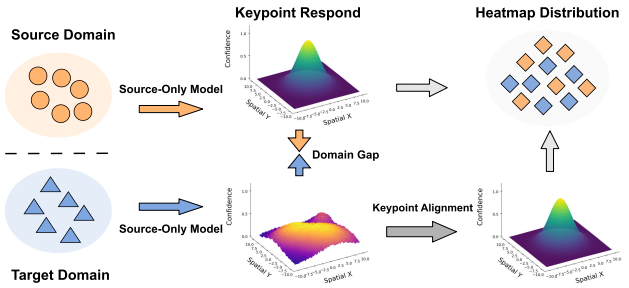


Fig. 2. Keypoint alignment between source and target domains.

and noisy heatmaps due to the domain shift and inherent speckle noise, degrading localization performance.

The core idea of KAM is to use adversarial learning to enforce a structural prior. It encourages the target-domain heatmaps to statistically mimic the clean, unimodal Gaussian-like distributions learned from the source domain. By aligning this statistical structure, we are not matching raw pixel values, but rather teaching the model to suppress SAR-specific noise and focus on the underlying, domain-invariant geometry of the ship targets. This process transfers the ability to generate high-quality, high-confidence localization signals, ensuring robust keypoint detection across domains.

The KAM is designed to address this issue. Instead of performing simple pixel-level feature matching, it directly aligns the spatial probability distribution of keypoints in the output space. Specifically, through methods like adversarial training, the model is encouraged to produce target-domain heatmaps that resemble the clean, compact, and locally unimodal Gaussian-like structures observed in the source domain. By doing so, the model learns to ignore domain-specific distractors (such as background texture and lighting variations) and consistently produces high-confidence responses at the object centers across different domains, thus achieving robust cross-domain keypoint localization.

This part of the loss function is

$$L_{ht} = -\sum_{i,u,v}\left[D_i \log\left({}_{ht}p_i^{u,v}\right) + (1-D_i)\log\left(1-{}_{ht}p_i^{u,v}\right)\right] \quad (1)$$

where ${}_{ht}p_i^{u,v}$ represents the domain discrimination prediction values at coordinates (u, v) of the ith image for the heatmap.

C. Box Alignment Module

To further reduce domain shift, AFS-Net complements the heatmap alignment with additional alignment on the offset and size maps.

1) Offset Alignment: By aligning the center point offset maps, the model can accurately correct the center locations of target regions, improving the alignment precision of local features. This approach reduces cross-domain discrepancies in keypoint prediction, resulting in higher alignment accuracy and greater task robustness. This part of the loss function is

$$L_{\text{off}} = -\sum_{i,u,v}\left[D_i \log\left({}_{\text{off}}p_i^{u,v}\right) + (1-D_i)\log\left(1-{}_{\text{off}}p_i^{u,v}\right)\right]. \quad (2)$$

2) Size Alignment: Aligning size maps improves cross-domain performance by reducing scale discrepancies between source and target domains, thereby enhancing BBox shape prediction, scale regression, and instance-level detection accuracy. This part of the loss function is

$$L_{wh} = -\sum_{i,u,v}\left[D_i \log\left({}_{wh}p_i^{u,v}\right) + (1-D_i)\log\left(1-{}_{wh}p_i^{u,v}\right)\right] \quad (3)$$

where ${}_{\text{off}}p_i^{u,v}$ and ${}_{wh}p_i^{u,v}$ represent the domain discrimination prediction values at coordinates (u, v) of the ith image for the offset map and size map, respectively.

D. Objective Function

The loss function of AFS-Net consists of two components: the object detection loss and the domain discrimination loss.

In the domain adaptation object detection task, ensuring the model's object detection performance is crucial. Since only the source domain is annotated, backpropagation relies exclusively on source-domain predictions. The object detection loss is defined as follows:

$$L_{\text{det}} = L_k + \lambda_{\text{size}} L_{\text{size}}^{\text{det}} + \lambda_{\text{off}} L_{\text{off}}^{\text{det}} \quad (4)$$

where L_k represents the heatmap loss, which is used to locate the center points of target objects; $L_{\text{size}}^{\text{det}}$ represents the size loss, which predicts the width and height of target objects; and $L_{\text{off}}^{\text{det}}$ represents the offset loss, which corrects the center point position deviation caused by downsampling. The parameters λ_{size} and λ_{off} are consistent with those in CenterNet.

In addition to the proposed keypoint and box alignment strategies, we incorporate a standard image-level alignment by applying a domain discriminator to the shared image features. This part of the loss is defined as follows:

$$L_{\text{img}} = -\sum_{i,u,v}\left[D_i \log\left(p_i^{u,v}\right) + (1-D_i)\log\left(1-p_i^{u,v}\right)\right] \quad (5)$$

where $p_i^{u,v}$ denotes the domain prediction at location (u, v) of the ith image.

The domain discrimination loss is defined as follows:

$$L_{da} = L_{\text{img}} + L_{ht} + L_{\text{off}} + L_{wh}. \quad (6)$$

Thus, the total objective function is defined as follows:

$$L = L_{\text{det}} + \lambda_{da} L_{da}. \quad (7)$$

The hyperparameter value $\lambda_{da} = 12$ is selected based on its best performance on the validation set.

The overall training process is shown in Algorithm 1. During the testing phase, only the CenterNet component is used, and the domain adaptation module is disabled.

III. Experiments

A. Datasets and Implementation Details

Experiments are conducted on three remote sensing datasets: LEVIR [15], HRSC [16], and SSDD [17]. LEVIR and HRSC are used as the labeled source domains, both containing only the "ship" class in our setting. The unlabeled target domain is SSDD. All datasets are split into training, validation, and test sets following a 7:1:2 ratio.

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Algorithm 1 Training Procedure of the Model

for training iterations **do**
  Sample mini-batch data x_i^S, x_i^T from $\mathcal{D}_S$, $\mathcal{D}_T$
  Extract image-level features $f_{img}(x_i^S)$, $f_{img}(x_i^T)$
  Generate heatmap, offset map, and size map from heads:
  $h_{ht}(x_i^S)$, $h_{ht}(x_i^T)$
  $h_{off}(x_i^S)$, $h_{off}(x_i^T)$
  $h_{wh}(x_i^S)$, $h_{wh}(x_i^T)$
  for each source heatmap, offset map and size map **do**
    Calculate the detection loss L_{det}
  end for
  for paired features $f_{img}(x_i)$, $h_{ht}(x_i)$, $h_{off}(x_i)$, $h_{wh}(x_i)$ **do**
    Calculate domain discriminant probabilities:
    $p_i^{(u,v)}$, ${}_{ht}p_i^{u,v}$, ${}_{off}p_i^{u,v}$, ${}_{wh}p_i^{u,v}$
    Calculate alignment losses:
    L_{img}, L_{ht}, L_{off}, L_{wh}
  end for
  Update network parameters by backpropagation
end for

TABLE I
PERFORMANCE COMPARISON WITH DIFFERENT METHODS

Method	Type	Backbone	LEVIR → SSDD	HRSC → SSDD
CenterNet-DLA34	anchor-free	DLA-34	38.83%	34.70%
CenterNet-R50	anchor-free	ResNet-50	23.90%	13.40%
DA	anchor-based	ResNet-50	40.74%	42.00%
SWDA	anchor-based	ResNet-50	44.30%	39.25%
CR-DA	anchor-based	ResNet-50	48.10%	39.20%
CR-SW	anchor-based	ResNet-50	42.92%	44.67%
IDA	anchor-based	ResNet-50	52.69%	44.30%
LRA-FCOS	anchor-free	VGG-16	56.60%	37.80%
LRA-Faster R-CNN	anchor-based	VGG-16	56.40%	40.40%
FIE-Net	anchor-free	VGG-16	50.99%	37.94%
AFS-Net-DLA34	anchor-free	DLA-34	55.20%	**46.80%**
AFS-Net-R50	anchor-free	ResNet-50	**57.27%**	45.20%

The models are implemented in PyTorch with ResNet-50 and DLA-34 backbones. The Adam optimizer is employed with a learning rate of 0.0005 and a batch size of 4 (two source and two target images). The code will be available at https://github.com/Lab-PANbin

B. Experimental Results

To validate the effectiveness of AFS-Net, we conducted comparative experiments with several state-of-the-art cross-domain object detection algorithms: DA [18], SW [19], CR [20], LRA [21], and FIE-Net on the same datasets. We evaluated the model using average precision (AP) at an IoU threshold of 0.5, a standard metric in object detection. Since inference in most domain adaptation methods depends only on the detector and backbone, inference efficiency is not compared here.

Table I summarizes the results, where the first column lists the methods, the second column specifies the corresponding backbone, and the third and fourth columns report the AP scores on the LEVIR → SSDD and HRSC → SSDD tasks, respectively.

On the LEVIR → SSDD task, AFS-Net establishes a new state of the art. With the ResNet-50 backbone, AFS-Net achieves a top performance of 57.27% AP, which is a remarkable 33.37% improvement over the CenterNet-R50 baseline. This result also surpasses all other competing domain adaptation methods by a significant margin of 0.67%–16.53%. When using the lightweight DLA-34 backbone, AFS-Net maintains highly competitive performance at 55.20% AP, demonstrating its strong adaptability.

For the HRSC → SSDD task, AFS-Net again shows strong performance. Notably, the lightweight DLA-34 variant achieves the best AP of 46.80%, outperforming the CenterNet-DLA34 baseline by 12.10%, and exceeding other domain adaptation methods by 2.13%–9.00%. Interestingly, it even surpasses the ResNet-50 variant (45.20%), suggesting that in scenarios with large appearance gaps, lightweight backbones like DLA-34 may generalize better by avoiding overfitting to source-specific features.

Overall, AFS-Net consistently establishes a new state of the art across both challenging domain adaptation tasks. It not only significantly surpasses baseline detectors but also outperforms recent, specialized domain adaptation methods.

TABLE II
ABLATION STUDY ON THE EFFECT OF DIFFERENT ALIGNMENT MODULES. ✓ INDICATES THAT THE COMPONENT IS USED

Method	KAM	BAM	LEVIR → SSDD	HRSC → SSDD
Baseline (w/o both)			50.37%	42.00%
+ BAM		✓	51.20%	43.18%
+ KAM	✓		53.20%	46.38%
Full (Ours)	✓	✓	**55.20%**	**46.80%**

C. Further Analysis

1) Ablation Study: Ablation studies are conducted to evaluate the contribution of different alignment components. Specifically, the CenterNet model combined with image-level alignment is adopted as the baseline, and the KAM and BAM are successively removed to assess their respective effectiveness. All experiments are conducted using DLA-34 as the backbone. The results are summarized in Table II.

The results show that both KAM and BAM improve performance. Adding KAM brings larger gains than BAM on both transfer tasks, highlighting the importance of keypoint alignment. Combining both modules yields the best results, confirming their complementary roles.

2) Visualization Experiment: Fig. 3 illustrates the qualitative detection results, comparing AFS-Net against CenterNet (source only) and the SWDA model on randomly selected test images.

To show the effectiveness of keypoint alignment, the heatmaps generated by AFS-Net are visualized alongside the original images and those produced by CenterNet (source only), as illustrated in Fig. 4. The comparative results reveal that AFS-Net achieves a more precise and distinct localization of ship keypoints. This indicates that the model successfully learns transferable keypoint representations from the source domain and effectively adapts them to the target domain.

IV. CONCLUSION

This letter proposes AFS-Net, an anchor-free, cross-domain ship detection framework based on the CenterNet architecture.

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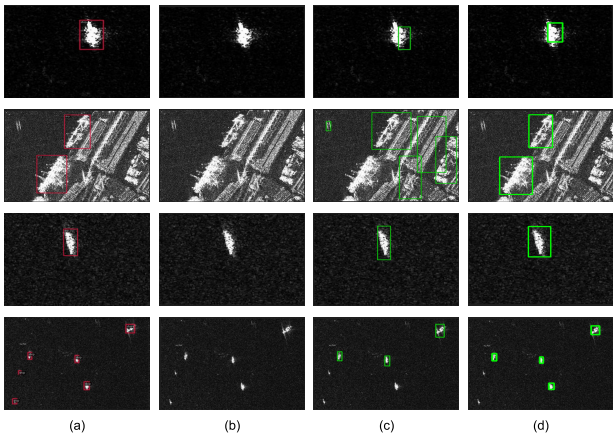


Fig. 3. Qualitative comparison of detection results on cross-domain tasks. The first two rows show the domain adaptation results from HRSC to SSDD, and the last two rows show the results from LEVIR to SSDD. (a) Ground truth. (b) CenterNet-R50 (source only). (c) SWDA. (d) AFS-Net.

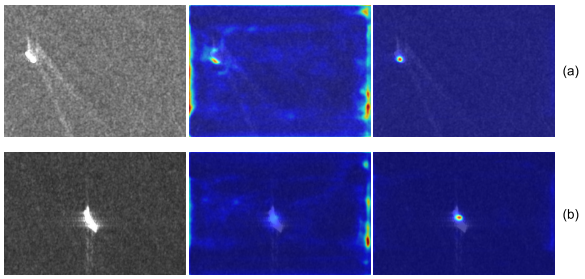


Fig. 4. Qualitative results of keypoint alignment. (a) and (b) Two representative examples. Each group displays: (left) input image, (center) heatmaps from source-only model with domain gap artifacts, and (right) aligned heatmaps from AFS-Net with improved localization and confidence.

By eliminating predefined anchor boxes, the model architecture is simplified and can better capture the structural characteristics of ships. We design two complementary modules for output-space alignment: the KAM and the BAM, which align the structural distribution of heatmaps and the geometric regression parameters of bounding boxes, respectively. This approach is suitable for unlabeled SAR ship detection scenarios and achieves superior experimental performance. However, some limitations remain, including sensitivity to hyperparameter selection. Future research will aim to develop more robust domain adaptation methods.

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